interpolate_st_mev_to_grid.R

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`interpolate_st_mev_to_grid` *Interpolate 3-D Spatiotemporal MEVs to Prediction Grid*

Description

Uses Inverse Distance Weighting (IDW, power = 2) in a z-scored 3-D space '(x_km, y_km, age_kyr)' to approximate spatiotemporal Moran Eigenvector Map (MEM) values from training samples to prediction locations at a given age. The interpolated values are then scaled using the training spatial scale attributes.

Usage

```
interpolate_st_mev_to_grid(
  data_st_mev_samples = NULL,
  data_coords_projected_train = NULL,
  data_coords_projected_pred = NULL,
  pred_age = NULL,
  spatial_scale_attributes = NULL
)
```

Arguments

`data_st_mev_samples`
: A data frame with row names in the format '"<dataset_name>__<age>"' and unscaled spatiotemporal MEV columns ('mev_1', ...), as returned by 'compute_spatiotemporal_mev()'.

`data_coords_projected_train`
: A data frame with 'coord_x_km' and 'coord_y_km' columns and 'dataset_name' as row names (site level), as returned by 'project_coords_to_metric()'.

`data_coords_projected_pred`
: A data frame with 'coord_x_km' and 'coord_y_km' columns and arbitrary row names identifying prediction locations, as returned by 'project_coords_to_metric()'.

`pred_age` Numeric or integer scalar. Age in years BP at which predictions are made. Used as the temporal dimension of the prediction 3-D coordinate matrix (‘age_kyr = pred_age / 1000‘).

`spatial_scale_attributes`
Optional named list of ‘"scaled:center"‘ and ‘"scaled:scale"‘ attributes per ST-MEV column, as returned by ‘scale_spatial_for_fit()‘ in the ‘spatial_scale_attributes‘ element. When ‘NULL‘, unscaled interpolated values are returned for fold-local scaling.

Details

3-D coordinates ‘(x_km, y_km, age_kyr)‘ of the training samples are z-scored using ‘colMeans‘ and column-wise standard deviations before computing Euclidean distances. This ensures the spatial and temporal extents contribute equally. The same z-score parameters are applied to the prediction grid coordinates, fixing the temporal dimension at ‘pred_age / 1000‘.

This function handles the **3-D spatiotemporal case**. For models fitted with ‘spatial_mode = "spatial"‘ use ‘interpolate_mev_to_grid()‘ instead.

Value

A data frame with the same row names as ‘data_coords_projected_pred‘ and one column per ST-MEV (names matching ‘data_st_mev_samples‘). Columns are scaled when ‘spatial_scale_attributes‘ is supplied and otherwise remain unscaled.

See Also

[compute_spatiotemporal_mev()], [interpolate_mev_to_grid()], [project_coords_to_metric()], [scale_spatial_for_fit()]

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